

INVENTION DISCLOSURE FORM

Confidential & Attorney-Client Privileged

SOORA NAMITHA

SR University

Applicant Address: Ananthasagar, Hasanparthy (M), Warangal Urban, Telangana - 506371,

1. Particulars of Inventors

Mr./Ms/Dr.	Name (Full)	Department	Designation	Mobile No.	Email	Postal Address
Mr.	Lade Gunakar Rao	CSE	Inventor1	9959320104	ladegunakarrao@gmail.com	+3-35/5,Bheemaram,Hanamakonda, Telangana- 506015
Ms.	Soora Namitha	CSE-AI&ML	Inventor2	9676335811	sooranamitha@gmail.com	1-125, Gopanapally,Parvathagiri, Warangal ,Telangana ,506365
Ms.	Challa Vyshnavi	CSE-AI&ML	Inventor3	8125013106	Vyshnavireddychalla13@gmail.com	1-24,kogilvai,Damera,Hanamkonda, Telangana,506006

2. Provide a brief descriptive title of the invention:

A: Machine Learning Based Health Insurance Cost Prediction Using Demographic and Health Data.

3. In 100 words or less, please provide an abstract or summary of the invention:

Abstract:

Health insurance cost prediction is an important task in the healthcare and insurance sector. Estimating medical insurance charges accurately is difficult because the cost depends on multiple factors such as age, gender, body mass index (BMI), smoking habits, and medical history. Traditional methods may not effectively analyze large and complex health datasets. This work proposes a machine learning-based framework that analyzes health and demographic data to predict insurance costs. The system uses data preprocessing, feature extraction, and regression algorithms to improve prediction accuracy. The proposed model learns patterns from historical data and provides reliable cost estimations, helping insurance companies and individuals make better financial and healthcare decisions.

4.Detail description of the in vention:(Answer to all below are required in detail)

a.Problem the invention is solving

The invention addresses the problem of accurately predicting health insurance costs based on various personal and health-related factors. Estimating insurance charges is a significant challenge in the healthcare and insurance industries because medical expenses depend on multiple variables such as age, gender, body mass index (BMI), smoking habits, medical history, and lifestyle patterns. Traditional estimation methods often rely on manual calculations or simple statistical approaches, which may not effectively analyze large healthcare datasets.

Current systems may produce inaccurate predictions or fail to capture complex relationships between different health factors and insurance charges. As a result, insurance companies may face difficulties in risk assessment, while individuals may not clearly understand their expected healthcare expenses.

Thus, the invention proposes a machine learning-based prediction system that can analyze large volumes of health and demographic data, learn patterns from historical records, and accurately estimate insurance costs. The system aims to support better decision-making for insurance providers and help individuals plan their healthcare finances more effectively.

b.General Utility/application of the invention

The invention has wide applications in the healthcare and insurance sectors where accurate prediction of medical insurance costs is important. It can be used by insurance companies to estimate insurance charges based on customer health and demographic data, helping them perform better risk assessment and design suitable insurance plans. The system can also assist customers in understanding their expected medical insurance expenses before purchasing a policy.

Additionally, the framework can be applied in healthcare analytics to analyze health-related factors such as age, body mass index (BMI), smoking habits, and lifestyle patterns that influence insurance costs. It can support hospitals, insurance providers, and policy planners in making data-driven decisions. Furthermore, the system can be integrated into online insurance platforms to provide instant cost predictions, improving transparency, efficiency, and accessibility in healthcare insurance services.

c. Advantages of the invention disclosing about the increased efficiency/efficacy

In the case of the project titled Machine Learning Based Health Insurance Cost Prediction Using Demographic and Health Data, the invention enhances efficiency and effectiveness in predicting costs of health insurance through the advanced machine learning algorithms. This, in comparison to conventional techniques, which use manual calculations and only take into consideration a few factors, is because this system will take into consideration numerous demographic and health-related factors, including age, body mass index (BMI), smoking status, number of dependents, and regional information to make more accurate predictions. The model is fast in processing large datasets and finds out complex associations between variables leading to faster and more dependable cost estimation. This will minimize human error, it will save time to the insurance providers and also it will be possible to give individualized premium to individuals. All in all, the system improves the efficiency of the decision-making of insurance companies and delivers reasonable and objective insurance cost forecasts.

d. Best way of using the invention as well as possible variants

The most appropriate approach to the invention is to apply the machine learning model to an electronic insurance management system in which demographic and health-related data including age, gender, body mass index, smoking habits, dependents and geographical data are given as input. This data is fed into the system and through trained machine learning algorithms, the system is able to accurately estimate the cost of health insurance of an individual. It assists insurance firms in making assumptions in a short time and make sound decisions using information. The potential versions of the invention are to apply other machine learning models like linear regression, decision trees, random forest or gradient boosting models to achieve better prediction results. As well, the system may be further enhanced by incorporating more health parameters, medical history, or real time health monitoring data to increase the performance and flexibility of prediction in various healthcare and insurance settings.

e. Working of invention along with Drawing, schematics and flow diagrams if required with complete explanations

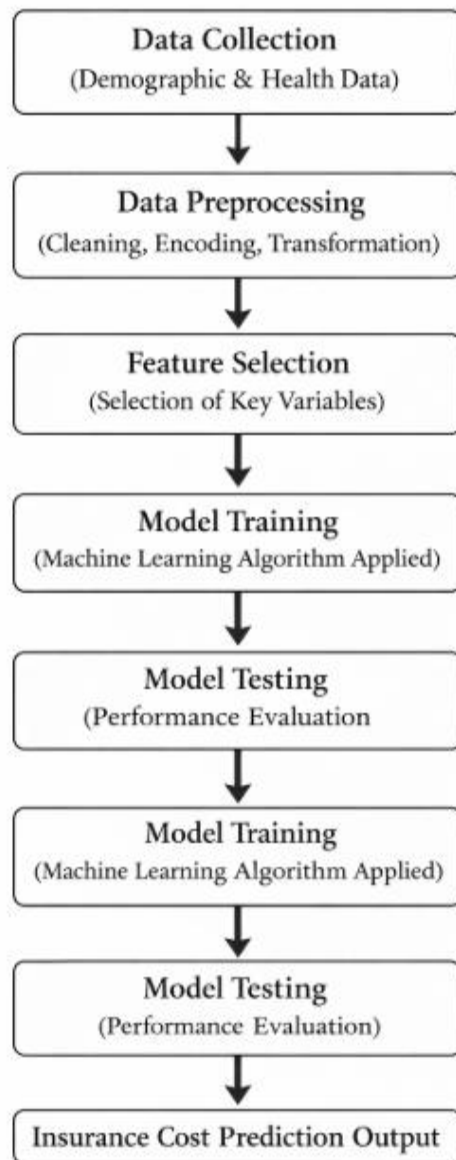


Figure 1: Flowchart of Machine Learning-Based Health Insurance Cost Prediction System

5. Have you conducted Primary Patent Search? Yes / No (if yes, attach the patent search report)

A: No

6. Existing state-of-the-art and prior arts: (Brief background of the existing knowledge/product/process in the market)

The current state of art in the health insurance sector primarily depends on the time-honored actuarial and statistical approaches to determine the insurance premiums. These mechanisms are based on the past medical records and preexisting risk profiles to estimate the cost of insurance depending on age, gender, body mass index (BMI), smoking status, the number of dependents, and location. The process in most cases is done manually with the insurance experts evaluating the risk and coming up with proper premium values. Nevertheless, those conventional approaches tend to be incapable of analyzing massive and intricate data and do not necessarily reveal latent connections between various health and demographic factors. Consequently, the predictions cannot always be very accurate and customized. As technology progressed, scholars have begun to investigate machine learning to enhance prediction accuracy and efficiency in health insurance cost estimation although most current systems remain limited due to limited data integration and poor scalability to emerging conditions in the field of health.

7. List out the known ways about how others have tried to solve the same or similar problems? Indicate the disadvantages of these approaches. In addition, please identify any prior art documentation or other material that explains or provides examples of such prior art efforts.

Researchers and insurance companies have been coming up with several ways of solving the health insurance cost prediction problem. Estimating of premiums based on past medical history and risk factors like age, gender, BMI, smoking, and medical history has traditionally been done with the help of actuarial models and statistical methods. These models are based on pre-set lines of assumptions and fixed formulae to calculate insurance premiums. Over the past few years, scholars have used machine learning methods, including linear regression, decision trees, random forest, gradient boosting, and artificial neural networks to enhance the accuracy of their predictions by focusing on huge volumes of data and finding patterns between health variables and insurance payments. Nevertheless, there are also some drawbacks of these approaches. Conventional statistical techniques do not always represent the complex relationship between variables and are likely to give inaccurate results. Conversely, other modern machine learning models can be large

and need to have high computing power, and can be black-box models that are challenging to understand. The use of machine learning to predict the medical insurance premium has been documented during the previous research studies, including such studies as, Medical Insurance Cost Prediction Using Machine Learning and Predicting Health Insurance Premiums Using Machine Learning, where the authors have explained various models that can be applied in estimating the insurance premiums based on the demographic and health data.

8. List the Technical features and Elements of the invention along with the Description of your invention from start to end.

The invention has to do with a machine learning-powered system that is supposed to estimate the price of health insurance and use demographic and health-related information to predict costs. The technicalities of the invention are the data collection, data preprocessing, feature selection, model training, prediction generation and the result output via a user interface. The system first takes the necessary input information which includes age, gender, body mass index (BMI), smoking status, number of dependents and the geographical area. This is followed by the preprocessing step where missing values are treated, inconsistent values are purged and categorical variables are transformed into numeric form to be analyzed. Upon preprocessing, imminent features that greatly determine insurance expenses are chosen and they are utilized to train the machine learning model. The system then uses appropriate algorithms like linear regression, decision tree or random forest that learns patterns on historical insurance data. After training and testing the model, the model is deployed to take in new user input and estimate the cost of health insurance. Lastly, the system interface displays to the user the predicted insurance premium. This has been achieved through this end-to-end process that allows the efficient, accurate and automated prediction of insurance costs, to enhance decision-making in the insurance providers and customers.

9. List out the features of your invention which are believed to be new and distinguish them over the closest technology.

The invention presents a machine learning-based health insurance cost prediction machine which has a few new features when compared to the current technologies. In comparison to the traditional actuary models which are based on the use of fixed statistical formulas and limited types of risks, this invention involves the use of advanced machine learning algorithms, which allow analysis of a wide range of demographics and health-related variables at once. The system can determine intricate interrelations among aspects like age, body mass index, smoking, number of dependents and regions to come up with improved, personalized estimates of insurance costs. The other distinguishing factor is the automated process of preprocessing data and feature selection which enhances the efficiency and reliability of the prediction model. Moreover, the system has the

ability to handle large volumes of data in a relatively short time and constantly enhance the accuracy of prediction, which was trained and evaluated through model training and evaluation. These characteristics make the invention unique among the existing means since it offers a more flexible, data-driven, and scalable means of health insurance cost prediction.

10.Has the invention been built or tested or implemented? If so, please provide the particulars of the first time it was successfully built or implemented (when, where, by whom, and evidence of this event including written or on-line pointers to documentary evidence):All testing details

Yes, the invention is developed and tested with the help of a machine learning framework of forecasting the cost of health insurance according to the demographic and health-related data. The system was applied with a publicly available medical insurance dataset that was retrieved on Kaggle. Its development and testing were done in the project implementation phase in the programming environment, say Python, along with machine learning libraries like Scikit-learn, Pandas, and NumPy. The dataset was separated into a training and testing dataset to test the performance of the prediction model. Various algorithms like linear regression, decision tree and random forest were used to do analysis of demographic variables like age, gender, BMI, smoking status, number of dependents, and region. The system was actually tested by comparing estimated insurance costs with reality insurance costs in the dataset. To calculate the performance based on performance measures like mean absolute error (MAE), mean squared error (MSE), and R2 score to find out the accuracy of prediction, the evaluation was conducted. The findings showed that machine learning models could be used to predict insurance costs with reasonable accuracy, which proves the viability and usefulness of the invention of automated cost estimation of health insurance.

11.Briefly state when and how you first conceived this idea?

This invention came to mind when examining and exploring the use of machine learning in machine learning systems in the health sector and insurance companies. During the process of discussing the ways demographic and health-related data can affect the medical costs, one realizes the necessity of a more efficient and precise way of estimating the costs of health insurance. The conventional insurance pricing procedures are largely based on the statistical and manual risk evaluation that might not be able to make individual predictions. Thus, the idea of applying the machine learning algorithms to a set of variables, including age, BMI, smoking status, number of dependents, and region, was created. The concept was developed in the course of a research work and project aimed at implementing data-based methods to enhance decision-making in the

healthcare insurance industry. The invention tries to combine both demographic and machine learning to become more specific, automated and efficient at the cost of health insurance predictors.

12. Have you sold, offered for sale, publicly used or published anything related to this invention? If yes, please briefly explain the dates and circumstances. List those individuals to whom you have revealed your invention. Were non-disclosure documents signed prior to disclosure in each case? Please state any deadlines of which you may be aware for filing an application on this invention.

No, the invention was not sold, proposed to sell, publicly used or distributed commercially. The idea is designed purely on academic and research grounds as a project pertaining to machine learning application in the prediction of health insurance costs. The academic supervisors, project guides, and team members in the research work have been involved only in discussing the details of the invention with the aim of guidance and evaluation. The invention was not officially announced in journals, conferences and any other public fields before this documentation was prepared. Moreover, there were no formal non-disclosure agreements (NDAs) that were signed prior to the discussion of the idea within the academic setting, considering that the discussion was restricted to the educational supervision. Currently, we are not aware of any particular times that are set aside as to the filing of an application regarding this invention, however it is hoped that any application pertaining to an intellectual property or patent should be made ahead of any disclosure or commercial exploitation of the invention.

13. Include any reasons that your invention would not have been obvious to someone of average skill in the art.

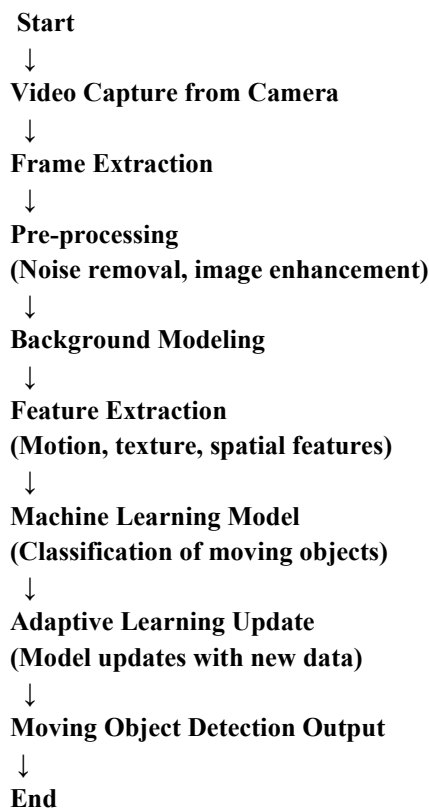
The invention would not have been obvious to a person having ordinary skill in the field because traditional health insurance cost estimation methods primarily rely on actuarial tables and simple statistical models that consider limited variables and predefined risk categories. In contrast, the proposed invention integrates machine learning techniques to analyze multiple demographic and health-related factors simultaneously, enabling the discovery of complex and nonlinear relationships between these variables and insurance costs. The combination of automated data preprocessing, feature selection, and the application of predictive algorithms provides a more accurate and adaptive system for estimating insurance premiums. Furthermore, the ability of the model to learn from large datasets and continuously improve prediction performance introduces a level of automation and intelligence that is not present in conventional approaches. Therefore the integration of these techniques in a unified system for personalized insurance cost prediction demonstrates a novel and non-obvious improvement over existing methods used in the industry.

14.Additional comments by inventor (if you want to give more details out of scope of this IDF).

The invention will enhance accuracy and reliability of the moving object detection in dynamic environment through the combination of adaptive learning techniques with the machine learning models. The suggested framework could be enhanced further with the help of the implementation of more sophisticated deep learning algorithms and more efficient feature extraction techniques to perform better.

The system can subsequently be expanded to facilitate real time intelligent surveillance system, smart city and vehicles and traffic management systems in the future. Additional studies and experimentation on a larger scale and on actual environments should assist in enhancing the efficiency, scale and the practicality of invention.

15.Drawings/Flowchart/Table



Form Submitted by:

Name : Soora Namitha

Contact No : 9676335811
